Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years of experience building scalable APIs and web systems. Skilled in Java/C++ OOP, SQL Server database design, and CI/CD, transitioning to C# and .NET.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Optimized **relational database schemas** and query patterns on **SQL Server**, reducing dashboard latency from **1.2s to 350ms** through index tuning and access pattern design for 500+ active users.
- Engineered automated validation checks and **ETL integration pipelines**, processing 5,000+ monthly jobs with 99%+ reliability while improving code quality and error logging to accelerate production issue resolution.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed robust full-stack features and **RESTful APIs** handling 10,000+ daily requests, translating complex product requirements into modular backend components and responsive user interfaces using React and Flask.
- Engineered **AI-assisted workflows** utilizing multimodal models and OCR, reducing feedback turnaround by **60%** while implementing automated software testing and validation methods to ensure production-level reliability across grading operations.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Designed large-scale data processing pipelines across **100,000+ trajectory records**, utilizing **Git** for version control to ensure reproducible benchmarking, cross-team collaboration, and high-confidence validation of autonomous agent behaviors.
- Prepared detailed technical validation reports and **automated unit test scenarios** for model-driven decision workflows, boosting closed-loop simulation stability by **15%** and improving cross-functional engineering alignment during system transition phases.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Optimized large-scale REST API integrations and data-sharing microservices on AWS using **Object-Oriented Programming (OOP)** principles and design patterns, resulting in a **30%** reduction in API response times.
- Developed and maintained **20+ automated test suites** using PyTest and Mockito in Jenkins CI/CD pipelines, boosting pipeline efficiency by **68%** while ensuring software quality and release reliability.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built full-stack web components and WebSocket-based real-time synchronization modules using the **MERN stack**, managing 2M+ synchronized records across locations with a **25%** reduction in code-review turnaround cycles.
- Designed and automated Docker-based build packaging and staging deployments using **Jenkins CI/CD pipelines**, cutting deployment time by **50%** and ensuring consistent software delivery across staging target environments.

Skills

Languages: C#, Java, C++, Python, TypeScript, JavaScript, SQL, HTML, CSS, Bash
Backend & Web: .NET, FastAPI, Flask, Node.js, Express.js, React, Next.js, REST APIs, WebSockets
Databases: SQL Server, PostgreSQL, MySQL, MongoDB, Relational Schema Design, Query Tuning
Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Git/GitHub
Engineering: Software Testing, Automation Testing, Agile/Scrum, OOP, System Design, Performance Tuning

Projects

Chat2SQL Code | *Java, TypeScript, SQL Server, Docker*
- Built an AI-powered database management workspace supporting **SQL Server** and 10+ database systems, utilizing **Java** and **TypeScript** to accelerate query authoring, schema exploration, and database structure synchronization.

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*
- Designed an AI-powered SaaS image platform with subscription-based billing via **Stripe**, scalable image storage in **MongoDB**, and a responsive **Next.js** interface, increasing user premium conversions by **18%**.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed containerized RESTful APIs on **AWS Fargate** using **Docker**, integrating automated **CI/CD pipelines** and validation testing gates to achieve production-grade service stability and code quality.